



IB Python API Lecture

Overview of EC2 Service on AWS

Running trades in the cloud

Before we begin, there are a few prerequisites as listed below. Some of us might already be aware of them. For those who are trying cloud services for the first time, let's go through each one by one.

Prerequisites

1. Basic understanding of cloud computing (e.g.: AWS).
2. RDP client.

Basic understanding of cloud computing (e.g.: AWS).

- Cloud computing
 - Cloud computing means that instead of all the computer hardware and software that we use being on our desktop, it's provided for us as a service by another company and accessed over the internet. It is just like using a remote computer.
 - For our session we are going to use Amazon Web Services (AWS).
- Amazon Web Services (AWS)
 - AWS is the most popular and widely used cloud service provider. AWS offers reliable, scalable, and inexpensive cloud computing services. Free to join, pay only for what we use.
 - AWS comes with many components and services. But we will focus only on the following ones.
 - AWS account creation and free tier.
 - AWS Management console.
 - EC2 and Types of EC2.
 - EC2 Launch / Start / Stop.
 - Security Groups and usage of security groups.

AWS account creation and free tier

- We can create a free account on AWS and get an advantage of the free tier given by AWS.
- AWS - Getting started -> <https://aws.amazon.com/getting-started/>
- AWS - Free Tier -> <https://aws.amazon.com/free/>

AWS Management console

- Access and manage Amazon Web Services through a simple and intuitive web-based user interface.
- Read more on -> <https://aws.amazon.com/console/>

EC2 and Types of EC2

- EC2
 - EC2 stands for Elastic Compute Cloud (EC2). It is a virtual machine (computer) which is customizable and scalable.
- Types of EC2
 - We can choose from various types of hardware configurations as per our need.
 - For example: t2.nano (1 CPU, 0.5GB RAM), t2.micro (1 CPU, 1GB RAM), t2.small (1 CPU, 2GB RAM), t2.medium (2CPU, 4GB RAM)
- Choosing a Type of EC2
 - For taking advantage of free tier, an EC2 instance type of t2.micro for 750 hours/month for 12 months can be a good choice.
 - Depending on our requirements we can opt for more hardware backed types of Ec2.
- Read more about types of EC2 and pricing
 - <https://aws.amazon.com/ec2>
 - <https://aws.amazon.com/ec2/instance-types/>
 - <https://aws.amazon.com/ec2/pricing/>

EC2 Launch / Start / Stop

- Amazon Machine Images (AMI)
 - AMI are like copies of machines (computers) which are ready for us to use.
 - AMI acts as a source from which we can launch (create) a number of machines as per our need.
 - Read more on AMI -> <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>
- Choosing AMI
 - We can choose from multiple AMI available on the AWS marketplace. For our session we will choose the AMI which we have already created with all the required software installed and ready to use.
- Launching an EC2 instance
 - To launch an EC2 instance we will make use of the AWS management console.
 - Simply click on EC2 dashboard and select Launch Instances and follow the steps.
 - Read more on launching EC2 -> <https://aws.amazon.com/ec2>
- Starting and shutting down EC2 instance
 - Once the EC2 is launched we can manage via the AWS management console (EC2 Dashboard).

Security Groups and usage of security groups

- Security Groups
 - Security groups act as a Firewall of your EC2.
- Usage
 - By default a security group is added during the launching of EC2 instances.
 - It consists of IP addresses which we want to allow to have access to our EC2 instance.
 - Security groups should be modified with care as it can allow unwanted access to our servers.
- Read more on AWS Security Groups
 - <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-network-security.html#ec2-classic-security-groups>